

No. of Printed Pages : 4
Roll No.

188844

Level 4, 2nd Sem / DVOC (Production Tech.)

Subject : General Mechanical Engineering - I

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

- Q.1 What is the ratio of stress to strain called?
a) Modulus of rigidity b) Modulus of elasticity
c) bulk modulus d) Poisson's ratio
- Q.2 Which type of load causes shear force and bending moment in a beam?
a) axial load b) transverse load
c) torsional load d) uniform load
- Q.3 Which type of gear is used to transmit power between non-parallel shafts?
a) Spur gear b) Bevel gear
c) helical gear d) rack and pinion
- Q.4 Bernoulli's theorem is based on the principle of conservation of _____
a) mass b) momentum
c) energy d) velocity

- Q.5 Which turbine is an impulse turbine?
a) Kaplan b) Francis
c) Pelton d) Propeller

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 Define the different types of pipe losses in fluid flow.
Q.7 What is the purpose of a bolted joint?
Q.8 What is the function of bevel gear in power transmission?
Q.9 Write two types of gears.
Q.10 What is the use of a Venturimeter?

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

- Q.11 Explain the relationship between shear force and bending moment.
Q.12 Describe different types of pulleys used in power transmission.

- Q.13 What is Bernoulli's theorem? Explain its application in fluid mechanics.

- Q.14 Explain the working of a reciprocating pump in brief.

- Q.15 Describe the different types of belt used in power transmission.

- Q.16 Define the various elastic constraints.

- Q.17 What are the different types of beams?

- Q.18 Define the different types of strain.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

- Q.19 Explain different types of riveted joint with neat sketches.

- Q.20 Describe in detail the working of a Pelton wheel turbine with a labeled diagram.